

ECON 724

Journal Review Presentation: The Lifetime Costs of Bad Health by
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① A Little Bit Of Background

② Health process

③ The Life-Cycle Model

④ Results

Main research question

- Healthy people earn more (income health gradient)
- Healthy people are wealthier (wealth health gradient)
- How costly it is to be unhealthy?
- What makes being unhealthy costly?
- To answer these questions we need to model health dynamics.
- How to model health dynamics?

Data

- Panel Study of income dynamics (PSID) 1984 to 2017
- Health and retirement study (HRS) 1992 to 2016 (cross check moment)
- both biannual
- Sample: Men with only high school education

Data

- Self reported health status
- not surprisingly
- higher chance of being in bad health when one get older
- Lower chance to revert bad health to good health when old
- higher chance to transition from good to bad health when old.



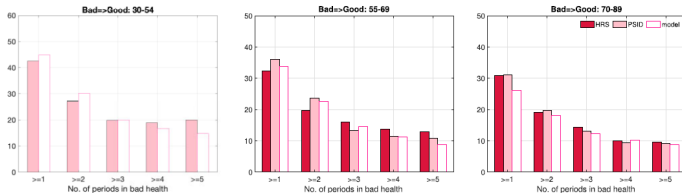
FIGURE 1

Figure: figure 1

Duration matters!

- negative duration dependence

(a)



(b)

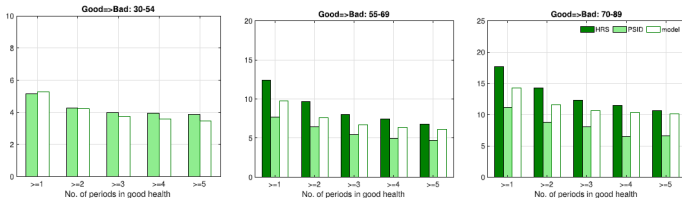


FIGURE 2

Figure 2

Data

- 2 mechanism for negative duration dependence
- compounding of health
 - modeled by higher order Markov chain.
- ex ante health difference (think genetic, lifestyle)
 - different transition probability for different health type
 - the model try to include both

fixed labor productivity and survival probability (estimation)

- Determine how fixed labor productivity related to health.
- Look at worker younger than 70, full time (520 hours/year) worker.
- $\log(inc_{it}) = \sum_{t=21}^{69} \sum_{j=\{G,B\}} d_t^j (D_{it}^{age})(D_{hit=j}) + \hat{\gamma}_i + u_{it}$
- Unobserved fixed labour productivity $\hat{\gamma}_i$, includes cohort effects in individual fixed productivity.
- Regress $\hat{\gamma}_i$ with on birth-year dummies to remove cohort effect, denoting the residual of this second regression estimates as γ_i .
- composition of fixed labor productivity are different for healthy and unhealthy in both PSID and HRS.

fixed labor productivity

- composition of fixed labor productivity are different for healthy and unhealthy in both PSID and HRS.
- To model this - allow correlation between fixed labor productivity type and health type

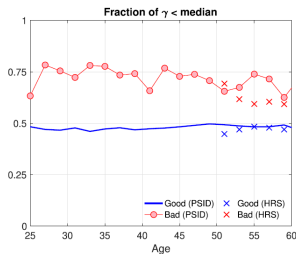


FIGURE 3

Figure: figure 3

constructing unobservable health types

- Use a ordered-logit model to assign/predict health types
- $$\Pr(\eta \leq \eta_j \mid \mathbf{X}_{t_0}^n) = \begin{cases} \Lambda(b_{\eta j} + \mathbf{B}_{\eta} \times \mathbf{X}_{t_0}^n) & \text{for } j = \{1, 2\}, \\ 1 & \text{for } j = 3. \end{cases}$$
- $\mathbf{X}$ contain fixed productivity tercile dummy $\gamma_L, \gamma_M, \gamma_H$, initial health h_{t_0} , wealth k_{t_0} , birth-year dummies.
- $\mathbf{B}$ is the coefficient for each dummy
- $b_{\eta j}$: constant term, $b_{\eta 1} < b_{\eta 2}$

Health process (the main contribution)

$$Pr(F_{t+1} \cup P_{t+1} | h_t, \tau_B, \eta) = \Lambda \left(\sum_{\tau=1}^{T-1} a_{\tau}^B \mathbb{1}_{(\tau_B=\tau)} + a_T^B \mathbb{1}_{(\tau_B \geq T)} \right. \\ \left. + f_{age}^{h_t}(t) + b_1 + a_{\eta}^B D_{\eta} \right) \quad (1)$$

$$Pr(F_{t+1} \cup P_{t+1} | G_t, \tau_G, \eta) = \Lambda \left(\sum_{\tau=1}^{T-1} a_{\tau}^G \mathbb{1}_{(\tau_G=\tau)} + a_T^G \mathbb{1}_{(\tau_G \geq T)} \right. \\ \left. + f_{age}^G(t) + b_2 + a_{\eta}^G D_{\eta} \right) \quad (2)$$

$$\zeta_t(h_t, \tau_B) = \Lambda \left(\sum_{\tau=1}^{T-1} a_{\tau}^{\zeta B} \mathbb{1}_{(\tau_B=\tau)} + a_T^{\zeta B} \mathbb{1}_{(\tau_B \geq T)} + f_{age}^{\zeta h_t}(t) \right) \quad (3)$$

$$\zeta_t(h_t, \tau_G) = \Lambda \left(\sum_{\tau=1}^{T-1} a_{\tau}^{\zeta G} \mathbb{1}_{(\tau_G=\tau)} + a_T^{\zeta G} \mathbb{1}_{(\tau_G \geq T)} + f_{age}^{\zeta h_t}(t) \right) \quad (4)$$

Transition probabilities of health status

- Combine transition probabilities of health status (equation 1-4)
- health type assignment CDF
- estimated survival probabilities
- estimated labour productivity fixed effect
- Bootstrap! $B = 10000$
- maximize log-likelihood
- $$L(\Theta) = \sum_{i=1}^N \log \left(\sum_{j=1}^3 \Pr(\eta_j \mid \mathbf{X}_{i,t_0}^n) \times \Pr(h_{i,t_0+\tau}, \dots, h_{i,t_0+J} \mid h_{i,t_0}, \dots, h_{i,t_0+\tau-1}, \eta_j) \right)$$
- where Θ is the parameters in equations 1,2,3,4, health type assignment function
- bootstrapping to get confidence interval

Transition probabilities of health status

- Final product of this health dynamic discussion

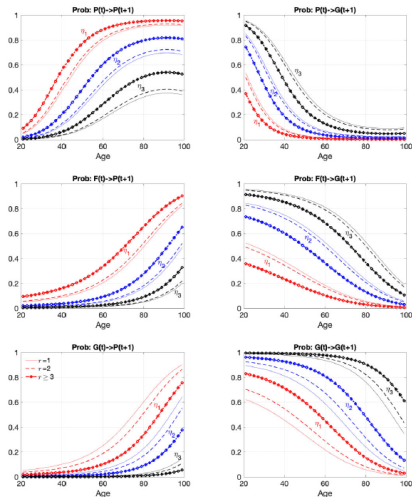


FIGURE 4
Estimated 2-year transition probabilities ($T = 2$)

Transition probabilities of health status

- Final product of this health dynamic discussion

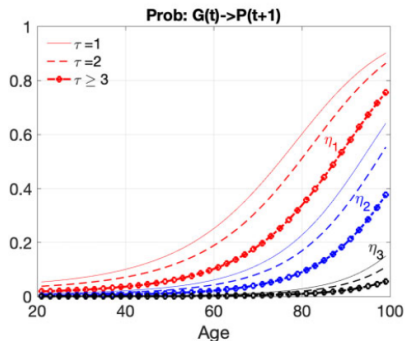


Figure: Prob($P \leftarrow G$)

Transition probabilities of health status

- Final product of this health dynamic discussion

TABLE 3

Top row: Initial distribution of individuals over health types at age 21. Other rows, joint distribution of $\{\eta, \gamma\}$ at age 21

	η_1	η_2	η_3
$Pr(\eta)$	0.08	0.35	0.57
$Pr(\eta \gamma_L)$	0.13	0.44	0.43
$Pr(\eta \gamma_M)$	0.08	0.36	0.56
$Pr(\eta \gamma_H)$	0.04	0.24	0.72

Notes: The terms γ_L , γ_M , and γ_H refer to the bottom, middle, and top terciles of the estimated fixed productivity distribution, respectively.

Figure: estimated joint distribution of η, γ

On the existence of health type

- They look at the HRS data
- People who are 55-56 years old (high school educated) and alive for the subsequent 10 years
- higher # of unhealthy periods correlate to smoking, higher BMI, lower parent education and longevity at age 55.

TABLE 4
*Characteristics of a balanced panel of people who are healthy at age 55-56 and alive for the subsequent 10 years,
 organised by the number of unhealthy periods between ages 57 and 66*

# Unhealthy periods	Individuals' characteristics (HRS)					Model	
	Smoking (%)	BMI	Father alive (%)	Mother alive (%)	Parents' educ (years)	η_1 (%)	η_3 (%)
0-1	22.6	27.9	21.6	48.4	10.1/10.5	0.1	78
2-3	27.1	29.5	21.5	50.4	9.2/9.9	3	12
4-5	44.4	29.8	16.1	36.5	8.4/9.2	25	2

Notes: BMI is the average body mass index. The first and second numbers in the education column refer to average education years of father and mother in each cell, respectively.

Figure: characteristics at age 55 and future health

On the existence of health type

- also related to genetics (score) for lower educational attainment, higher chance to smoke, higher BMI and lower longevity.

TABLE 5

Average polygenic scores of people who are healthy at age 55–56 and alive for the subsequent 10 years, organised by the number of unhealthy periods between ages 57 and 66

# Unhealthy periods	Polygenic scores (HRS)			
	Educational attainment	Smoking	BMI	Longevity
0–1	−0.120	0.003	−0.006	−0.06
2–3	−0.216	0.023	0.127	−0.065
4–5	−0.708	0.092	0.140	−0.250

Figure: polygenic score and future health

Endogenous health does not matter

- The health process will break if health can be influenced by decisions.
- Medical expenses do not improve health or mortality
 - Danesh et al (2024)
 - Free healthcare in Netherland
 - high income people still live much longer
 - channel: chronic diseases correlated to early age income.
 - No significant different in mortality of those with chronic disease across income group.
 - low income group develop chronic disease faster and the difference increases with age.
 - Another evidence for ex-ante health difference!

Endogenous health does not matter

Exercise is overrated

- Include exercise, re-estimate ordered logistics
 - existing coefficient: same size and significance
 - exercise only plays a preventative role
 - reduce prob of bad health only if someone is currently healthy and fair
 - size of coefficient: health type dominate exercise

How health directly affects economic outcomes

- Two direct mechanisms at work.
- Incorporate the direct health effects into the structural model.
- Medical spending.
- Productivity.
- Disutility from work.
- Access to health insurance.
- Survival probability.

How individuals differ Ex-Ante Characteristics

- Fixed labour productivity $\gamma \in \{\gamma_L, \gamma_M, \gamma_H\}$, diecretized into three terciles.
- Health type $\eta \in \{\eta_1, \eta_2, \eta_3\}$.
- Patience (Discount factor) $\beta \in \{\beta_{low}, \beta_{high}\}, \beta_{low} = 0.877 < \beta_{high} = 0.992$.
- $3 \times 3 \times 2 = 18$. 18 Total number of heterogeneity sources in the model!
- However, these characteristics can be correlated with each other.
- Moment matching $\Pr(\beta_{low} | \eta_j) = \begin{cases} 0.778 & \text{for } j = 1 \\ 0.794 & \text{for } j = 2 \\ 0.38 & \text{for } j = 3. \end{cases}$
- Controlling for one's health type β, γ are independent.

Preferences

- Each period T is two years, retirement set at period R , live at most T periods.
- Endowed with 1 unit of time.
- Labour supply l_t is indivisible, $l_t \in \{0, 1\}$
- Fixed cost from working, and working when under poor or fair health, ϕ_W, ϕ_F, ϕ_P .
- An additional positive term to ensure that individuals value their life (i.e. sick people don't get utility from dying faster), $\bar{b}$.
- ρ is risk aversion and $\bar{n}_t$ is an age-specific household size.
- $u(c_t, l_t, h_t) = \frac{(c_t/\bar{n}_t)^{1-\rho}}{1-\rho} - \phi_W \mathbb{1}_{\{l_t > 0\}} - \phi_F \mathbb{1}_{\{h_t=F, l_t > 0\}} - \phi_P \mathbb{1}_{\{h_t=P, l_t > 0\}} + \bar{b}$
- Moment matching $\bar{b} = 7.149$
- Set outside $\rho = 3.0, \bar{n}_t 1.9 - 3.2$

Bequests

- $v(k) = \theta_{Beq} \frac{(k+k_{Beq})^{1-\rho}}{1-\rho}$
- De Nardi (2004).
- We have seen this Bequest model many times, k_{Beq} determines to what extent bequest are a luxury good, and θ_{Beq} determines the strength of the bequest motive.
- Homogenous bequests motives, however this will be affected by agents' ex-ante differences in characteristics.
- Moment matching $\theta_{Beq} = 1,905, k_{Beq} = 182,707$

Labour Productivity process

- Labour income $z_t^h l_t$
- Labour productivity $z_t^h = \lambda_t^h \Upsilon_t = \lambda_t^h \exp(v_t) \exp(\gamma)$,
- $v_t = \rho_v v_{t-1} + \varepsilon_t; \varepsilon_t \sim N(0, \sigma_\varepsilon^2 = 0.0198), \rho_v = 0.9472$
- $\gamma \sim N(0, \sigma_\gamma^2 = 0.051)$.
- If you ignore the fixed productivity γ component inside the idiosyncratic component Υ_t and take the log of the labour productivity, this is how we whack TFP using an AR(1) process to generate a real business cycle in DSGE.
- λ_t^h is deterministic function of age and current health

Medical expenses and health insurance

- At the beginning Medical expense shock x_t^h realizes. x_t^h follows the distribution of medical shocks $\mathcal{G}_t(x_t^h|h_t)$.
- Four types of health insurance: uninsured, individual insurance, employer-sponsored health insurance(ESHI), and Medicare. $i_H \in \{0, 1, 2, 3\}$.
- $cvg(x_t^h, i_H)$ as the fraction of medical expenses covered by insurance.
- May purchase individual insurance at market price $p_I(h_t, t) = \xi EM_t(h_t, t) + \varphi^h$. φ increases as health status worsen - insurance denial, higher search cost (100 or 2100).
- $\xi = 1.07$
- $P_{MCR} = 1,055$

Taxation and social transfers

- Medicare tax τ_{MCR} , Social Security tax τ_{ss} , consumption tax τ_c and tax on capital income τ_k .
- Progressive labour income tax $\mathcal{T}(y) = y - a_{\tau 0} y^{1-a_{\tau 1}}$, Heathcote et al. (2020)
- Public safety-net program, $T^{SI}(\bar{c}) = \max\{0, (1 + \tau_c)\bar{c} + Tax + P_t^h + x_t^h(1 - cvg(x_t^h, i_H)) - k_t(1 + (1 - \tau_k)r) - z_t^h l_t\}$.
- Social Security: $ss(h_{R-1}, v_{R-1}, \gamma, \eta, \beta)$.
- Moment matching $\bar{c} = 3,505$
- Set outside $a_{\tau 0} = 0.941, a_{\tau 1} = 0.158, \tau_k = 0.05, \tau_c = 0.36$

Timing of the models

- State variables are: $k_t, h_t \in \{G, F, P\}$, $\tau \in \{1, 2, 3\}$, $v_t, g_t \in \{0, 1\}$, $t \in \{1, 2, \dots, R-1\}$, $\gamma \in \{\gamma_L, \gamma_M, \gamma_H\}$, $\eta \in \{\eta_1, \eta_2, \eta_3\}$, $\beta \in \{\beta_{low}, \beta_{high}\}$. Based on this information, an individual decides labour supply l_t , and insurance choice i_H .
- At the beginning of the period, individuals learn their productivity, health and ESHI offer status. At the end of each period, x_h^t shock is realized, after paying the out-of-pocket medical expenses individual then choose c_t and k_{t+1}
- control variables for working-age individual are: l_t, i_H, c_t, k_{t+1}
- control variables for retirees are: c_t, k_{t+1}
- Denote $\mathbb{S}_t = [k_t, h_t, \tau, v_t, g_t, \gamma, \eta, \beta]$

Optimization problem

- Now let's add everything together.
- For working-age individuals ($t < R$)

$$V_t(\mathbb{S}_t) = \max_{l_t, i_H} \sum_{x_t^h} \mathcal{G}_t(x_t^h | h_t) W_t(\mathbb{S}_t; l_t, i_H, x_t^h) \quad (5)$$

- where we have interim value function

$$\begin{aligned} W_t(\mathbb{S}_t; l_t, i_H, x_t^h) = \max_{c_t, k_{t+1}} \{ & u(c_t, l_t, h_t) + \beta [\zeta_t^h E_t(V_{t+1}(\mathbb{S}_{t+1})) \\ & + (1 - \zeta_t^h) \theta_{Beq} \frac{(k + k_{Beq})^{1-\rho}}{1 - \rho}] \} \end{aligned} \quad (6)$$

Optimization problem

- subject to

$$k_t(1 + (1 - \tau_k)r) + z_t^h l_t - x_t^h(1 - cvg(x_t^h, i_H)) - P_t^h - Tax + T^{SI}(\bar{c}) = (1 - \tau_c)c_t + k_{t+1} \quad (7)$$

$$T^{SI}(\bar{c}) = \max\{0, (1 + \tau_c)\bar{c} + Tax + P_t^h + x_t^h(1 - cvg(x_t^h, i_H)) - k_t(1 + (1 - \tau_k)r) - z_t^h l_t\} \quad (8)$$

$$Tax = \mathcal{T}(z_t^h l_t) + \tau_{MCR} z_t^h l_t + \tau_{ss} \min\{z_t^h l_t, \bar{y}_{ss}\} \quad (9)$$

- Insurance premium: $P_t^h = \begin{cases} 0 & i_H \in \{0, 2\} \\ p_I(h_t, t) & i_H \in \{1\} \end{cases}$

Optimization problem

- For retirees ($t \geq R$)

$$V_t(\mathbb{S}_t^R) = \max_{l_t, i_H} \sum_{x_t^h} \mathcal{G}_t(x_t^h | h_t) W_t(\mathbb{S}_t^R; x_t^h) \quad (10)$$

- where we have interim value function

$$\begin{aligned} W_t(\mathbb{S}_t^R; x_t^h) = \max_{c_t, k_{t+1}} \{ & u(c_t, 0, h_t) + \beta [\zeta_t^h E_t(V_{t+1}(\mathbb{S}_{t+1}^R)) \\ & + (1 - \zeta_t^h) \theta_{Beq} \frac{(k + k_{Beq})^{1-\rho}}{1 - \rho}] \} \end{aligned} \quad (11)$$

Optimization problem

- subject to

$$k_t(1 + (1 - \tau_k)r) + ss - x_t^h(1 - cvg(x_t^h, 3)) - P_{MCR} - \mathcal{T}(ss) + T^{SI}(\bar{c}) = (1 - \tau_c)c_t + k_{t+1} \quad (12)$$

$$T^{SI}(\bar{c}) = \max\{0, (1 + \tau_c)\bar{c} + \mathcal{T}(ss) + P_{MCR} + x_t^h(1 - cvg(x_t^h, 3)) - k_t(1 + (1 - \tau_k)r) - ss\} \quad (13)$$

Results

- Compositional differences and wealth-health gradient
- monetary losses
- welfare losses

Compositional differences and wealth-health gradient

- Calibration result: time preference and health type is correlated

TABLE 8
Preference parameters and the consumption floor

Parameters		Baseline	No correlation
(Annual) discount factor	$\{\beta_{low}, \beta_{high}\}$	{0.877, 0.992}	{0.895, 0.992}
% β_{low} by η_i at age 20	$Pr(\beta_{low} \eta_1)$	78%	55.6%
	$Pr(\beta_{low} \eta_2)$	79%	55.6%
	$Pr(\beta_{low} \eta_3)$	38%	55.6%
Bequest parameter	θ_{Beq}	1,905	1,256
	κ_{Beq}	\$182,707	\$168,577
(Annual) consumption floor	$\bar{c}$	\$3,505	\$4,116

Notes: Both β and $\bar{c}$ are converted into annual values.

Figure: second stage estimated parameter

Results

- Compositional differences and wealth-health gradient
- No-correlation model can not replicate wealth-health gradient

TABLE 12
Wealth-health gradient at age 60–64 (in 1,000 USD)

Wealth difference by health status	PSID (HRS)	Baseline	No correlation $Pr(\beta_{low} \eta_i) = 0.556$
25th pct	\$56 (\$47)	\$67	\$58
50th pct	\$142 (\$98)	\$146	\$35
75th pct	\$210 (\$222)	\$260	\$99

Notes: The table reports the wealth difference between healthy and unhealthy (poor+fair) people for each wealth quartile.

Figure: wealth-health gradient near retirement

- we need correlation between ex-ante heterogeneity.

monetary losses

- monetary loss to bad health
- counterfactual experiment (no bad health realization)

TABLE 13

Annual monetary losses due to bad health (poor + fair)

	All	η_1	η_2	η_3
Over life-cycle (21-death)				
Percentage of time in bad health	15%	58%	23%	4%
Income losses + total medical costs	\$1,511	\$8,896	\$1,935	\$225
(Percentage of average earnings)	(3.9%)	(23%)	(5%)	(0.6%)
Over working age (21–64)				
Percentage of time in bad health	10%	55%	14%	1%
Income losses + total medical costs	\$1,031	\$7,147	\$1,201	\$76
(Percentage of average earnings)	(2.7%)	(18%)	(3%)	(0.2%)

Notes: The top panel is over life-cycle till death while the bottom panel is between 21 and 64 (working ages). The interest rate for computing the present value is 2%. Average earning in our baseline model is \$38,648 per year.

Figure: annual monetary losses

monetary losses

TABLE 14

Composition of annual monetary loss due to bad health (poor + fair) using 2% interest rate when computing the present value

	Over life-cycle (21-death)				Over working periods (21-64)			
	All	η_1	η_2	η_3	All	η_1	η_2	η_3
Annual monetary losses	\$1,511	\$8,896	\$1,935	\$225	\$1,031	\$7,147	\$1,201	\$76
Composition (%)								
Medical costs paid by insurance	36	33	39	39	32	33	33	18
Out-of-pocket medical costs	27	22	30	36	20	20	21	11
Income losses	37	45	31	24	48	47	46	71

Figure: Composition of annual monetary losses

welfare losses

compensating consumption equivalent (CCE) approach

TABLE 16
Welfare losses due to bad health (poor + fair)

	all	η_1	η_2	η_3	β_L	β_H
Compensated consumption equivalence	\$1,933	\$6,380	\$2,690	\$854	\$1,866	\$2,018
(% consumption equivalence, λ_c)	(10.6)	(36.8)	(14.8)	(4.4)	(10.3)	(11)
Contribution (%)						
Only medical expenses losses	25	39	22	17	24	26
Only income losses	38	57	42	9	55	17
Only non-monetary losses	44	32	33	77	14	79

Notes: The dollar value is calculated from $\lambda_c \bar{c}^{**}$ where λ_c and $\bar{c}^{**}$ are the percentage of consumption reduction and average life time consumption of each individual when always healthy.

Figure: annual welfare losses due to bad health

welfare losses

TABLE 17

Concentration of losses due to bad health (poor + fair) and variation of losses due to health types

	Over life-cycle (21-death)			Variation due to η (%)
	Top 5%	Top 10%	Top 20%	
Monetary losses (21-death)				
Income losses + total medical costs	38%	56%	75%	69
Welfare losses				
Compensated consumption equivalence	24%	42%	71%	30

Figure: Concentration of losses